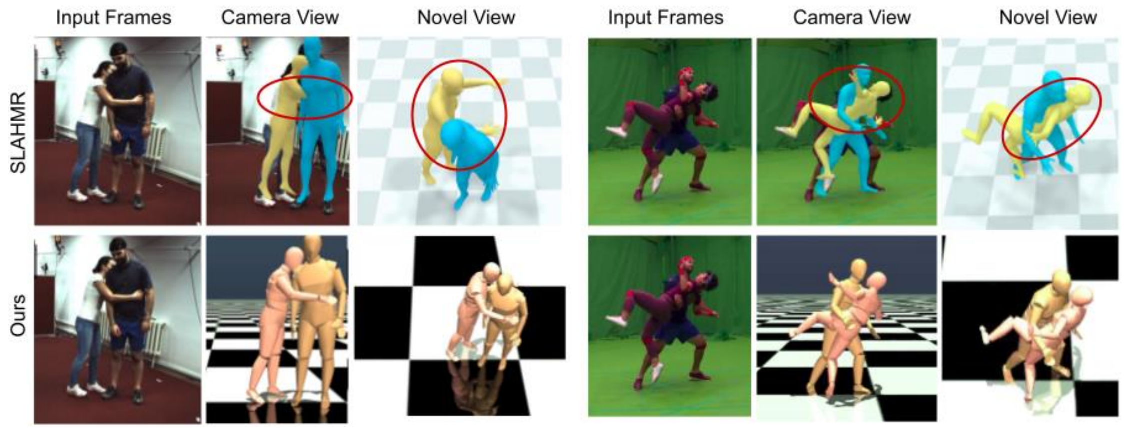
# Tracking

The Problem

Current methods track people well,

**but the results aren't physical.**

- Feet slide on the ground
- Bodies go through floors
- Motion looks jittery
- Can't use it in a physics simulator




MultiPhys (CVPR'24) — adding physics fixes artifacts

**Fix: Add physics constraints. Track the person and the scene together. Make it physically realistic**

# Tracking

Recent Methods

**Body trackers (video → 3D body):**

GVHMR (SigAsia'24) — best accuracy, fast

Human3R (arxiv'25) — real-time, multi-person

**Scene reconstruction (video → 3D scene):**

Depth Anything 3 — depth + geometry from video

CUT3R — underlies Human3R

**Physics-aware methods:**

MultiPhys (CVPR'24) — physics simulator loop

PhysHMR (SigAsia'25) — end-to-end physics

**No paper has combined all three with real scene geometry yet.**

# Tracking

## Proposed Approach

**Input: Video**

↓

**Body Tracker**

GVHMR → 3D body poses
- GVHMR directly gives world frame that physics needs. Not multi-person. ~5000 fps

**Scene Reconstruction**

Depth Anything 3 → 3D geometry
- DA3 gives metric depth (real world meters), 78 FPS, and code has been released

↓ ↓

### Physics Refinement (Optimization-based)

1.Scene proxy: Ground plane estimation from DA3 depth + collision mesh

2.Contact detection: Foot vertices near ground + low velocity → contact

3. Optimize SMPL params to minimize: penetration, foot skating, jitter, instability *[PROX'19, LEMO'21, IPMAN'23]*

4. Why not RL: PhysHMR (SigAsia'25) showed RL post-processing worsens foot sliding (5.65→12.71mm) due to corrective limb movements

**Output: Physically realistic 3D motion + scene**

# Literature Review (Body Tracking)

# Tracking

Literature Review

## SAM-Body4D

arxiv 2025

- Training-free 4D human body mesh recovery from videos
- Builds on SAM 3D Body (per-frame) but adds temporal consistency without any retraining
- Uses a video segmentation model to generate identity-consistent masks across frames
- Occlusion-Aware module restores body regions hidden behind objects or other people
- Refined masks guide SAM 3D Body to produce temporally coherent mesh sequences
- Padding-based parallel strategy for efficient multi-person processing
- Handles occlusion and multi-person scenarios that per-frame methods struggle with

**Multi-person tracker, training-free approach. No world frame**

# Tracking

Literature Review

## GVHMR

SIGGRAPH Asia 2024

- Recovers 3D body motion in world coordinates from a single video
- Defines a "gravity-view" coordinate system — uses gravity direction + camera angle to place the person in the real world
- Estimates each frame independently, then a transformer refines the sequence — avoids error accumulation
- Gets camera rotation from visual odometry or phone gyroscope
- State-of-the-art: PA-MPJPE 36.2mm (3DPW), WA-MPJPE 111.0mm (EMDB-2)
- Very fast: ~5000 FPS for the core network
- Single-person only, no scene reconstruction

**Good for single-person body tracking — best accuracy and speed**

# Tracking

Literature Review

## WHAM

CVPR 2024

- Recovers global 3D human motion from video using learned motion context
- Recurrent architecture that integrates visual + motion features over time
- Can optionally use phone IMU data (accelerometer/gyroscope) for better trajectory
- Fast: ~5 seconds per 1000 frames
- Good accuracy (PA-MPJPE 35.9mm) but more foot sliding than GVHMR (4.4mm vs 3.0mm)
- Single-person only

**Strong alternative to GVHMR — slightly worse on foot sliding, comparable otherwise**

# Tracking

Literature Review

## Human3R

arxiv 2025

- Recovers multi-person 3D bodies and the scene simultaneously from a stream of images
- Built on CUT3R (scene) + Multi-HMR (bodies), fine-tuned together
- "One model, one stage, trained in one day on one GPU"
- Real-time: 15 FPS, 8 GB VRAM
- Handles multiple people, unknown cameras, varying layouts
- Authors note limitations: body-scene penetration and no human-object interaction
- **Thought: limitations can be fixed through physics?**

**Multi-person**

# Literature Review (Scene Reconstruction)

# Tracking

Literature Review

## ZipMap

CVPR 2026

- 3D scene reconstruction from images with linear time complexity (not quadratic like DUSt3R/VGGT)
- Uses "test-time training layers" to compress an entire image collection into a compact hidden scene state in a single forward pass
- Processes images sequentially while maintaining a compressed state — no need to compare all image pairs
- Over 700 frames in under 10 seconds on a single H100 — 20x faster than VGGT
- Matches or exceeds reconstruction quality of quadratic-time methods
- Supports streaming reconstruction

**Faster and more scalable alternative to VGGT/DUSt3R. Could be a good option**

# Tracking

Literature Review

## JOSH

ICLR 2026

- Jointly optimizes 3D bodies and scene reconstruction from video
- Multi-stage: estimates per-frame, then jointly optimizes with contact + scene consistency losses
- Multi-person capable
- Best offline accuracy: WA-MPJPE 68.9mm — much better than real-time methods
- Tradeoff: offline optimization, not real-time
- Shows that jointly reasoning about humans and scenes improves both

**Best results overall but too slow for real-time — proves joint human+scene reasoning helps**

# Tracking

Literature Review

## Depth Anything 3

arxiv 2025

- Estimates metric depth, camera poses, and 3D point clouds from video
- "Depth-ray" representation: predicts depth + camera ray direction per pixel. Recovers camera parameters without needing them as input
- Streaming: processes frames one at a time, works on any video length
- Memory efficient: < 12 GB VRAM
- 78 FPS

**Good for scene geometry — streams any-length video, gives us depth + camera poses for the physics layer**

# Tracking

Literature Review

## CUT3R

CVPR 2025

- Reconstructs 3D scenes from image sequences in a streaming, online fashion
- Successor to DUSt3R (CVPR'24), which needed all image pairs at once (quadratic cost)
- CUT3R processes frames one at a time — much more scalable
- Outputs dense point maps, depth, and camera poses
- 8 GB VRAM, works on dynamic scenes
- Forms the backbone of Human3R — Human3R adds human mesh recovery on top

**What Human3R is built on**

# Literature Review (Physics)

# Tracking

Literature Review

## MultiPhys

CVPR 2024

- Takes existing body motion estimates and refines them using MuJoCo (physics simulator)
- Each person simulated as a separate articulated body — collisions resolved by physics engine
- Trains an RL policy to control each body so it tracks the original motion while obeying physics
- Results: 7x less body-ground penetration
- First method to handle multi-person physics-aware correction
- Limitation: RL policy can create artifacts (arm flailing for balance recovery)

**Physics simulation dramatically reduces artifacts — but the RL approach can backfire**

# Tracking

Literature Review

## PhysHMR

SIGGRAPH Asia 2025

- End-to-end: goes directly from video to physically plausible 3D motion
- Key finding: two-stage pipelines (track then correct) can make things worse
- GVHMR alone: 5.65mm foot sliding
- GVHMR + RL post-correction: 12.71mm foot sliding **(2.2x worse)**
- PhysHMR end-to-end: 4.60mm foot sliding (best)
- Physics awareness must be integrated, not bolted on as a post-process

**Proves physics helps accuracy — and warns that RL post-correction is the wrong approach**

# Tracking

Literature Review

## CRISP

arxiv 2025

- Combines physics simulation with scene-aware tracking
- Uses a planar scene representation (flat ground plane) + RL-based body control
- 8x lower failure rate compared to methods without scene awareness
- Even a simple flat ground plane significantly helps physics-based tracking
- Suggests richer scene geometry (like from DA3) could help even more

**Even a flat ground plane helps a lot — motivates using real scene geometry from DA3**

# PROX

ICCV 2019

- Pioneering work: use scene constraints to improve body pose estimation
- Given a 3D scene scan, optimizes body parameters to avoid penetrating the scene
- Uses signed distance fields (SDFs) to measure body-scene penetration
- 24% improvement in vertex-to-vertex accuracy with scene constraints
- Limitation: requires a pre-scanned 3D scene — not applicable to in-the-wild video alone

**The original proof that scene constraints improve body estimation — the optimization style we'd follow**

# Tracking

Literature Review

## LEMO

ICCV 2021

- Adds physics-inspired losses to smooth and correct body motion sequences
- Friction loss: feet shouldn't slide when touching the ground
- Temporal smoothness: penalize sudden jerky changes
- Ground contact consistency
- Eliminates foot skating and jitter — no physics simulator needed, just loss terms
- Optimization-based, simple, proven

**Our main reference for the optimization approach — simple loss terms, no simulator, proven results**

# Tracking

Literature Review

## PhysDiff

ICCV 2023

- Integrates physics into a diffusion model for human motion generation
- Applies physics-based corrections during each denoising step
- 86% reduction in physical errors vs. non-physics diffusion models
- Different domain (motion generation, not tracking)
- Shows physics constraints help even in very different architectures

**Different domain but further evidence: adding physics consistently improves results**